

Eldridge Surianto

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

Bachelor of Science in Computer Engineering | Minor in Applications of AI/ML | GPA: 3.95

Expected Graduation: May 2027

- **Concentrations:** Systems and Architecture & Computing Hardware and Emerging Architectures
- **Related Coursework:** Computer Architecture, VLSI & Advanced Digital Design, Physical Foundations of Computer Engineering, Computer Organization, Digital Design Laboratory, Data Structures & Algorithms, Web Design, Intro to Signal Processing

SKILLS

Programming: C/C++, Python, Rust, Verilog, VHDL, MIPS, MATLAB, Java, HTML, CSS, Javascript

Technical: FPGAs, HLS, IC Design, Data Structures & Algorithms, Finite Element Analysis, Web Development, Machine Vision

Tools: Cadence Virtuoso, Vitis HLS, Vivado, ModelSim, Quartus Prime, GTKWave, Linux CLI, Git, Django, MATLAB, COMSOL Multiphysics

Platforms: Linux, Windows, MacOS

Languages: English, Chinese, German

PROFESSIONAL EXPERIENCE

Research Software Engineer | LightningX | Sharc Lab | Georgia Tech

August 2025 - Present

- Refactored the waveform tracing backend for a Rust-based High-Level Synthesis (HLS) simulation framework, resulting in a 50% reduction in overall simulation execution time by implementing a waveform output module that drastically shrank file size and storage overhead, enabling more efficient debugging and reverse engineering of complex parallel and pipelined HLS designs.

Software Intern | Automation Technology Center PTE LTD | Singapore

May 2025 - August 2025

- Built a high-performance data processing pipeline, utilizing OpenCV for laser line image processing, OpenGL for interactive 3D visualization, and multithreading to reduce processing time for 500 images from over 60 seconds to less than 2 seconds; leveraged OpenMP for parallel computations, accelerating machine vision capabilities crucial for semiconductor manufacturing.
- Implemented a precision 3D laser scanning system for automated semiconductor lead frame inspection, enabling highly detailed and repeatable dimensional data capture by integrating EtherCAT (SOEM protocol) with a Yaskawa SGD7S servo drive for real-time encoder-triggered, low-level camera control via Spinnaker SDK acquisitions. The program has been successfully integrated into an industrial semiconductor device fabrication machine.

Undergraduate Research Assistant | Plasma & Dielectrics Lab | Georgia Tech

August 2024 - Present

- Designed MATLAB scripting pipelines for modeling zinc oxide voltage-dependent resistor (MOV) microstructures in COMSOL Multiphysics and assigning semiconductor granular properties, reducing Finite Element Analysis (FEA) setup time by over 99%.
- Executed thousands of FEA simulations using the Monte Carlo method to analyze the impact of material inhomogeneity on current distribution, generating statistical insights to optimize MOV design by identifying key MOV parameters controlled during manufacturing process affecting performance for high-energy handling power electronics.

PROJECTS

VHDL RTL LED Metronome on DE10-Standard ModelSim-Intel SoC FPGA

- Designed and implemented a VHDL-based LED metronome peripheral on a DE10-Standard FPGA using Quartus Prime and comprehensive ModelSim timing simulations, incorporating a state machine mechanism to manage the precise and visually engaging timing through PWM-controlled LED brightness and clock division for adjustable BPM.
- Integrated the peripheral into a custom SCOMP (Simple Computer) architecture, enabling dynamic tempo and LED control via SCOMP assembly instructions directly interfacing with memory-mapped registers.

Wordle Game Embedded Systems Programming on Mbed uLCD kit

- Developed a Wordle-inspired game in C++ for an ARM Cortex-M 32-bit microcontroller, achieving smooth real-time gameplay by optimizing memory usage and code efficiency on a resource-constrained system using GPIO and interrupts.
- Integrated dynamic visuals, audio, and responsive input handling by applying real-time debugging techniques and precise timing control, improving user responsiveness and strengthening hardware-software interaction across all gameplay states.

LEADERSHIP EXPERIENCE

ECE Discovery Studio Peer Leader Fellowship | Georgia Tech

August 2025 - Present

- Mentored a group of 15 students in an extended orientation course for over 200 first-year ECE students, providing guidance on academic and social opportunities at Georgia Tech. Facilitated class discussions and activities on professional development topics like resume writing, portfolio websites, elevator pitches, imposter syndrome, and growth mindset.
- Managed course logistics by taking attendance, encouraging discussion, and grading assignments such as resumes and course registration plans. Led student project showcase meetings, including a technical article analysis and a personal ECE project.

Product Owner | CS2340 Objects and Design | Georgia Tech

August 2024 - December 2024

- Led a 5-person team, strategically translating project requirements into actionable tasks and allocating responsibilities, while overseeing UI/UX and backend implementation to create Django web apps like a Food Finder and Spotify Wrapped app.
- Applied SCRUM Agile principles to manage member progress and focus on user-centric features in an engaging format.